www.tntextbooks.in

does not bind O_2. Normally RBC contains less than 1% methaemoglobin.

6.5 Transport of gases

6.5.1 Transport of oxygen

Molecular oxygen is carried in blood in two ways: bound to haemoglobin within the red blood cells and dissolved in plasma. Oxygen is poorly soluble in water, so only 3% of the oxygen is transported in the dissolved form. 97% of oxygen binds with haemoglobin in a reversible manner to form oxyhaemoglobin (HbO_2). The rate at which haemoglobin binds with O_2 is regulated by the partial pressure of O_2. Each haemoglobin carries maximum of four molecules of oxygen. In the alveoli high pO_2, low pCO_2, low temperature and less H^+ concentration, favours the formation of oxyhaemoglobin, whereas in the tissues low pO2, high pCO_2, high H^+ and high temperature favours the dissociation of oxygen from oxyhaemoglobin.

A sigmoid curve (S–shaped) is obtained when percentage saturation of haemoglobin with oxygen is plotted against pO_2. This curve is called oxygenhaemoglobin dissociation curve (Figure 6.7). This S–shaped curve has a steep slope for pO_2 values between 10 and 50mmHg and then flattens between 70 and 100 mm Hg.

Under normal physiological conditions, every 100mL of oxygenated blood can deliver about 5mL of O_2 to the tissues.

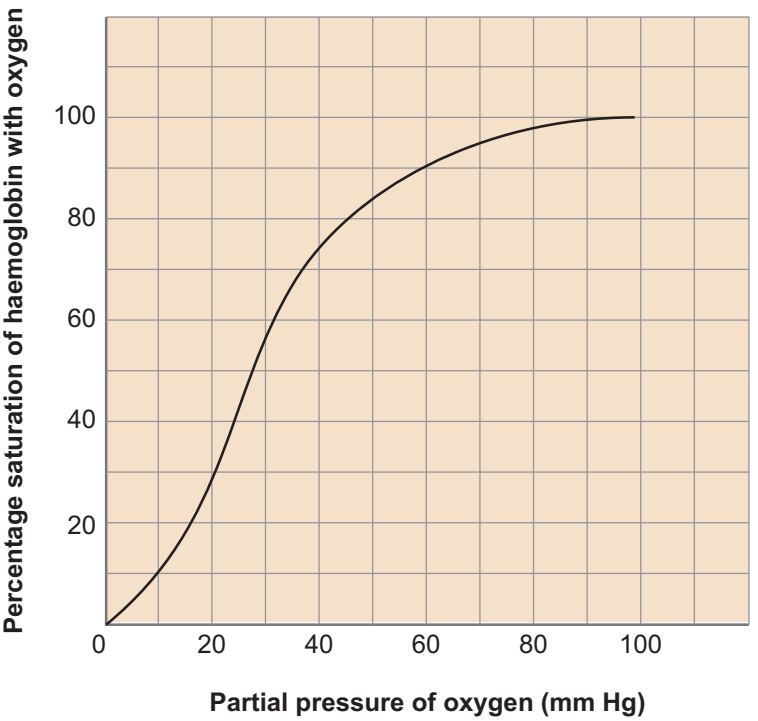

Figure 6.7 Oxygen dissociation curve

6.5.2 Transport of Carbon–dioxide

Blood transports CO_2 from the tissue cells to the lungs in three ways

i. **Dissolved in plasma** About 7 – 10% of CO_2 is transported in a dissolved form in the plasma.

ii. **Bound to haemoglobin** About 20 – 25% of dissolved CO_2 is bound and carried in the RBCs as carbaminohaemoglobin (Hb CO_2)

$CO_2 + Hb \rightleftharpoons Hb\ CO_2$

iii. **As bicarbonate ions in plasma** about 70% of CO_2 is transported as bicarbonate ions

This is influenced by pCO_2 and the degree of haemoglobin oxygenation. RBCs contain a high concentration of the enzyme, carbonic anhydras, Whereas small amounts of carbonic anhydrase is present in the plasma.

At the tissues the pCO_2 is high due to catabolism and diffuses into the blood to form HCO_3^- and H^+